

Data Fairness Reflection

1. Basis

To determine whether a neighborhood was underrepresented, I compared the number of unique members and the number of checkouts across all neighborhoods. I also calculated the percentage of members represented by each neighborhood. A neighborhood was considered potentially underrepresented if it had noticeably fewer members and checkouts than the other neighborhoods.

2. Finding

Shubra appeared to be underrepresented in the dataset. It had only 5 unique members, representing approximately 7.7% of the 65 members with known neighborhood information. It also had the lowest number of checkouts, with 34. In comparison, the other neighborhoods had between 11 and 20 members and between 68 and 114 checkouts.

Therefore, based on both member representation and checkout activity, Shubra was the least represented neighborhood in this dataset.

3. A Plausible Reason

One possible explanation is that fewer library members from Shubra were included in this data. Another possibility is that the library's services or summer programs reached residents of other neighborhoods more effectively. However, the dataset alone cannot confirm the exact reason for the difference.

4. A Next Step

Next summer, the library could increase outreach efforts in Shubra, such as promoting library programs and membership opportunities through local schools or community spaces. Afterward, the library could compare participation and checkout data again to see whether representation from Shubra improves.

A comparison table of regional vitality based on the number of members and the number of checkouts

neighborhood	members	checkouts	member_percentage
Heliopolis	13	87	20.000000
Maadi	20	114	30.769231
Nasr City	16	101	24.615385
Shubra	5	34	7.692308
Zamalek	11	68	16.923077